SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

Question 1: Online Symposium Portal

A college is developing an online symposium registration portal. The following form is observed:

```
<form action="/register" method="POST">\n  <input type="text" name="name">\n  <input type="email"
name="email">\n  <button>Submit</button>\n</form>
```

1. Identify appropriate HTML5 semantic elements missing in the structure.
2. Modify the structure to make it standards-compliant.
3. Interpret the HTTP request generated during form submission.
4. Analyze the HTTP response cycle from server to browser.
5. Suggest improvements for usability and accessibility.

Question 2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>\n  <head><title>Company</title></head>\n  <body>\n    <div>\n      <h1>Welcome\n      <p>About
us\n    </div>\n  </body>\n</html>
```

1. Identify improper nesting and missing closing tags.
2. Analyze how different browsers may interpret this structure.
3. Identify missing semantic elements (e.g., header, section).
4. Provide a corrected W3C-compliant HTML structure.
5. Suggest techniques to ensure cross-browser compatibility.

Question 3: Form Submission Failure

A form is not submitting as expected:

```
<form action="/submit" method="GET">\n  <input type="text" name="username">\n  <input type="password"
name="password">\n  <button type="button">Login</button>\n</form>
```

1. Identify issues in the form configuration.
2. Analyze why the HTTP request is not triggered.
3. Interpret the role of GET method in this scenario.
4. Propose a corrected implementation.
5. Suggest improvements for secure data transmission.

SOLUTION – QUESTION 1: ONLINE SYMPOSIUM PORTAL

1. Missing semantic elements

The original form lacks a clear page structure. A <header> can contain the page heading, <main> identifies the primary content, <section> groups the registration form, and <footer> contains supporting information. Each input should have a connected label.

2. Complete working HTML implementation

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Symposium Registration</title>
<style>
*{box-sizing:border-box}body{margin:0;min-height:100vh;display:grid;place-items:center;padding:20px;font-
family:Arial;background:linear-gradient(135deg,#312e81,#8b5cf6,#ddd6fe)}
.card{background:white;width:400px;max-width:100%;padding:30px;border-radius:22px;box-shadow:0 20px 50px
#17104755}
.tag{color:#7c3aed;font-size:12px;font-weight:bold;letter-spacing:2px}h1{color:#6d28d9}
p{color:#64748b;line-height:1.6}label{display:block;margin:15px 0 6px}
input{width:100%;padding:12px;border:1px solid #ddd6fe;border-radius:9px}
button{width:100%;padding:13px;margin-top:22px;border:0;border-
radius:9px;background:#6d28d9;color:white;font-weight:bold;cursor:pointer}
button:hover{background:#4c1d95}#success{display:none}a{color:#6d28d9}
</style></head><body>
<header></header><main class="card" id="registration"><span class="tag">SIMATS EVENTS</span>
<h1>Ideas Into Impact ✦</h1><p>Join our annual symposium and connect with future innovators.</p>
<form id="form"><label for="name">Full Name</label><input id="name" required placeholder="Enter your name">
<label for="email">Email Address</label><input id="email" type="email" required
placeholder="you@example.com">
<button type="submit">Register Now →</button></form></main>
<main class="card" id="success" aria-live="polite"><h1>Registration Complete! ✓</h1><p id="message"></p><a
href="" onclick="location.reload()">Register another student</a></main>
<footer></footer>
<script>
document.getElementById('form').addEventListener('submit',function(e){e.preventDefault();
document.getElementById('message').textContent='Thank you, '+document.getElementById('name').value+'!
Registration received for '+document.getElementById('email').value+'.';
document.getElementById('registration').style.display='none';document.getElementById('success').style.displa
y='block';});
</script></body></html>
```

3. HTTP request

When the form is submitted to a server using POST, the browser sends the field values in the request body with HTTP headers. The HTML-only version above demonstrates the interface locally and prevents navigation; it does not send a real POST request.

4. HTTP response cycle

- The browser sends the request to the server.
- The server validates and processes the registration.
- The server returns an HTTP status code, headers and response body.
- The browser displays the confirmation page or an error response.

5. Usability and accessibility

- Use connected labels, required fields and email input validation.
- Provide clear success and error messages.
- Ensure keyboard navigation, visible focus and sufficient contrast.
- Validate input on the server for a real deployment.

Screenshot

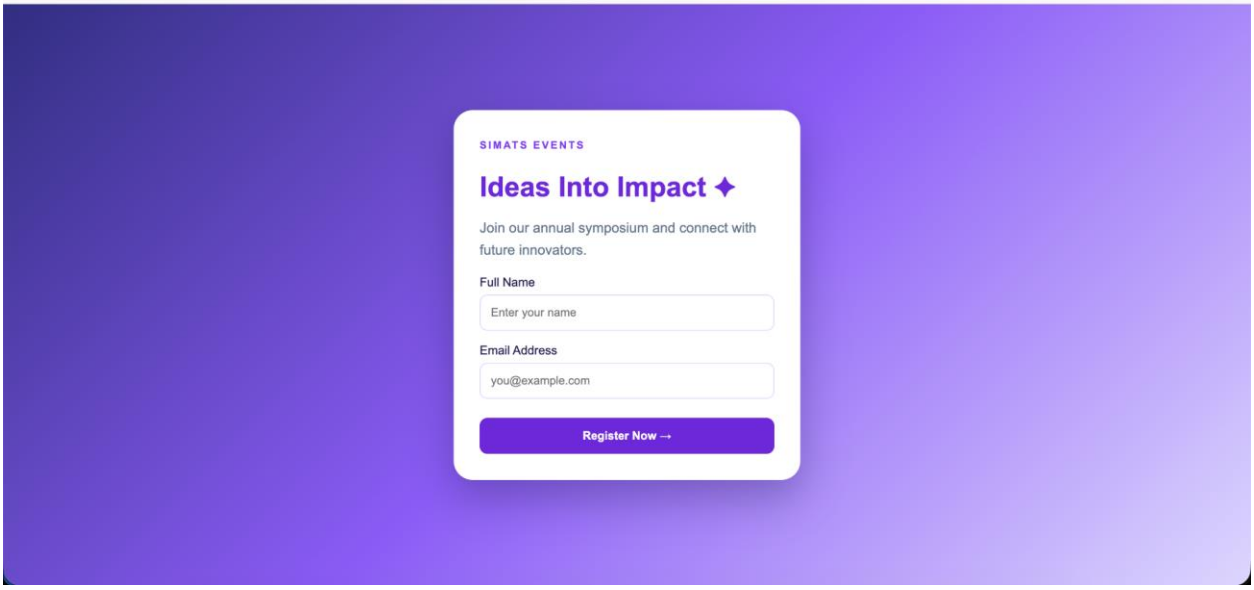


Figure 1. Symposium registration page.

SOLUTION – QUESTION 2: CROSS-BROWSER COMPATIBILITY

1. Improper structure

The original snippet omits <!DOCTYPE html> and the lang attribute, and it leaves heading and paragraph tags unclosed. Browsers may repair malformed markup, but the resulting document tree can be unpredictable.

2. Browser interpretation

Browsers use HTML parsing rules and may automatically close or insert elements. Differences in rendering engines and default styles can affect layout. Valid HTML and explicit CSS reduce these differences.

3. Semantic elements

Use <header> for the page heading, <nav> for navigation, <main> for primary content, <section> for grouped content, <article> for independent service cards, and <footer> for supporting details.

4. Complete working HTML implementation

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Nova | Digital Solutions</title>
<style>
*{box-sizing:border-box;scroll-behavior:smooth}body{margin:0;font-
family:Arial;background:#f4f7fb;color:#172554;line-height:1.6}
header,footer{background:#101c3a;color:white;padding:20px 7%}header{display:flex;justify-content:space-
between;align-items:center;flex-wrap:wrap;gap:12px}
.logo{color:#67e8f9;font-size:24px;font-weight:bold}nav{display:flex;gap:18px;flex-
wrap:wrap}a{color:inherit;text-decoration:none}nav a:hover{color:#67e8f9}
main{max-width:1050px;margin:auto;padding:25px}.hero{padding:45px;background:linear-
gradient(120deg,#172554,#2563eb);color:white;border-radius:18px}
.hero p{color:#dbeafe;line-height:1.7}button{cursor:pointer;border:0;border-radius:8px;padding:11px
15px;font-weight:bold}
.primary{background:#67e8f9;color:#172554}.grid{display:grid;grid-template-columns:repeat(auto-
fit,minmax(200px,1fr));gap:16px}
section{margin:24px 0}article,.contact{background:white;padding:22px;border-radius:13px;box-shadow:0 5px
20px #17255412}
article
h3{color:#2563eb}.details{display:none;color:#475569}input,textarea{width:100%;padding:12px;margin:7px 0
14px;border:1px solid #cbd5e1;border-radius:8px;font:inherit}
footer{text-align:center}.dark{background:#0f172a;color:#e2e8f0}.dark article,.dark
.contact{background:#1e293b;color:#e2e8f0}
.dark .details{color:#cbd5e1}@media(max-
width:550px){main{padding:15px}.hero{padding:25px}header{padding:18px}}
</style></head><body>
<header><div class="logo">NOVA.</div><nav aria-label="Main navigation"><a href="#home">Home</a><a
href="#about">About</a><a href="#services">Services</a><a href="#contact">Contact</a></nav>
<button onclick="document.body.classList.toggle('dark')">☾ Theme</button></header>
<main><section class="hero" id="home"><p>THINK BIG. BUILD BETTER.</p><h1>Digital Ideas, Made Real.</h1>
<p>Discover creative technology solutions that help your ideas grow.</p><button class="primary"
onclick="document.getElementById('services').scrollIntoView()">Explore Services →</button></section>
<section id="about"><h2>Who We Are</h2><p>Nova creates accessible websites and useful digital experiences
with modern technology.</p></section>
<section id="services"><h2>Our Services</h2><div class="grid">
<article><h3>✦ Web Design</h3><p>Beautiful responsive websites.</p><button onclick="show('web')">Learn
More</button><p class="details" id="web">We create accessible layouts for mobile, tablet and
desktop.</p></article>
<article><h3>⌘Development</h3><p>Practical web applications.</p><button onclick="show('dev')">Learn
More</button><p class="details" id="dev">We build useful applications with reliable
functionality.</p></article>
<article><h3>↗ Innovation</h3><p>Ideas for business growth.</p><button onclick="show('ideas')">Learn
More</button><p class="details" id="ideas">We explore technology to improve user experiences.</p></article>
</div></section>
<section id="contact" class="contact"><h2>Contact Us</h2><form id="contactForm"><label
for="name">Name</label><input id="name" required>
<label for="email">Email</label><input id="email" type="email" required><label
for="msg">Message</label><textarea id="msg" rows="3" required></textarea>
<button class="primary">Send Message</button></form><p id="notice" aria-live="polite"></p></section></main>
<footer><p>© 2026 Nova Digital Solutions</p></footer>
<script>
function show(id){let
e=document.getElementById(id);e.style.display=e.style.display==='block'?'none':'block'}
document.getElementById('contactForm').addEventListener('submit',function(e){e.preventDefault();document.get
ElementById('notice').textContent='Thank you, '+document.getElementById('name').value+'! Your message is
ready.';this.reset()});
</script></body></html>
```

5. Cross-browser compatibility techniques

- Use valid HTML and CSS and validate markup with W3C tools.
- Include DOCTYPE, lang and viewport metadata.
- Use responsive layouts and explicit CSS rather than browser defaults.
- Test in Chrome, Firefox, Safari and at different viewport sizes.
- Provide alternatives for unsupported features and keep keyboard accessibility.

Screenshot

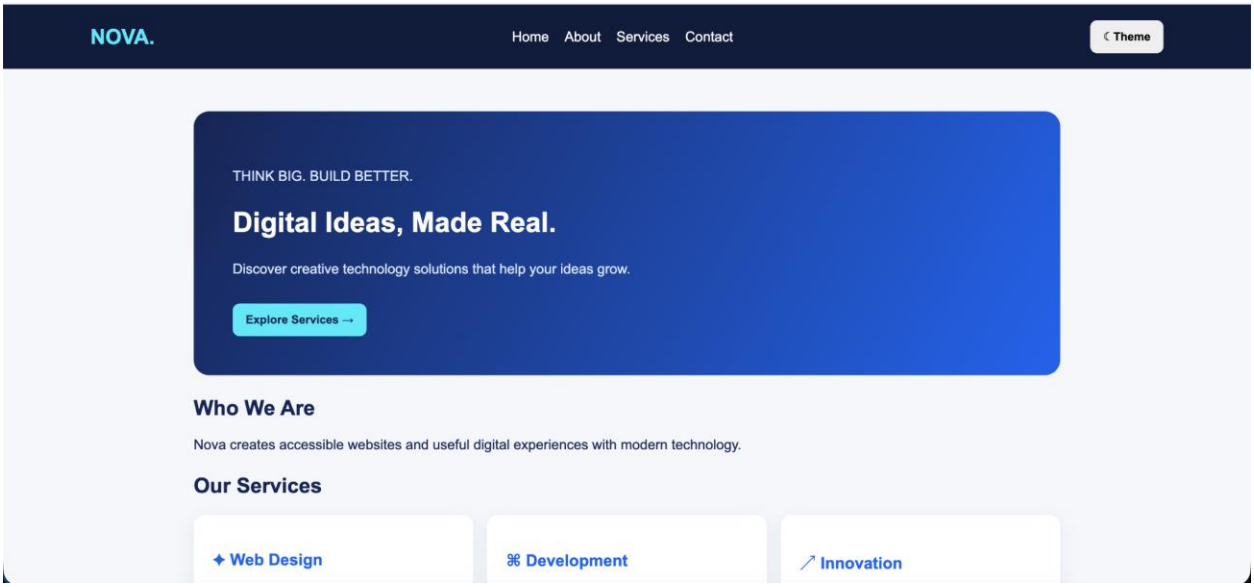


Figure 2. Nova website showing the responsive company layout.

SOLUTION – QUESTION 3: FORM SUBMISSION FAILURE

1. Issues in the configuration

The button uses type="button", so it does not submit the form automatically. GET appends form values to the URL and is inappropriate for passwords. A submit button and POST method are suitable for a login form.

2. Why the request is not triggered

A button with type="button" does not submit the form unless JavaScript explicitly handles submission. Using type="submit" starts the browser form submission process.

3. Role of GET

GET is intended mainly to retrieve resources. Form values are included in the URL, where they may appear in browser history or logs. Use POST over HTTPS for credentials; POST alone does not encrypt data.

4. Complete HTML implementation

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Elegant Login Form</title>
<style>
*{box-sizing:border-box}body{margin:0;min-height:100vh;display:grid;place-items:center;padding:20px;font-
family:Arial;background:linear-gradient(135deg,#e0e7ff,#f5f3ff,#d1fae5)}
.card{width:400px;max-width:100%;padding:35px;background:white;border-radius:22px;box-shadow:0 15px 40px
#64748b30}
h1{color:#5145cd}p{color:#64748b;line-height:1.5}label{display:block;margin:16px 0 7px}
input{width:100%;padding:12px;border:1px solid #dbe1ed;border-radius:9px}
button{width:100%;margin-top:22px;padding:13px;border:0;border-
radius:9px;color:white;background:#635bdb;cursor:pointer}
a{color:#5145cd;text-decoration:none}#success{display:none}
</style></head><body>
<main class="card" id="login"><h1>Welcome back ✦</h1><p>Sign in to continue your journey.</p>
<form id="loginForm"><label for="user">Username</label><input id="user" required placeholder="Your
username">
<label for="pass">Password</label><input id="pass" type="password" required placeholder="Your password">
<button type="submit">Sign In →</button></form></main>
<main class="card" id="success" aria-live="polite"><h1>Success! ✓</h1><p id="message"></p><a href=""
onclick="location.reload()">← Back to login</a></main>
<script>
document.getElementById('loginForm').addEventListener('submit',function(e){e.preventDefault();
document.getElementById('message').textContent='Form submitted successfully. Welcome,
'+document.getElementById('user').value+'!';
document.getElementById('login').style.display='none';document.getElementById('success').style.display='bloc
k'});
</script></body></html>
```

5. Security improvements

- Use HTTPS with a valid TLS certificate.
- Use POST for credentials; POST alone does not encrypt data.
- Validate input on the server and never store plain-text passwords.
- Use secure session cookies, salted password hashes and rate limiting in a real application.
- Disable debug mode in production.

Screenshots: Login and successful submission

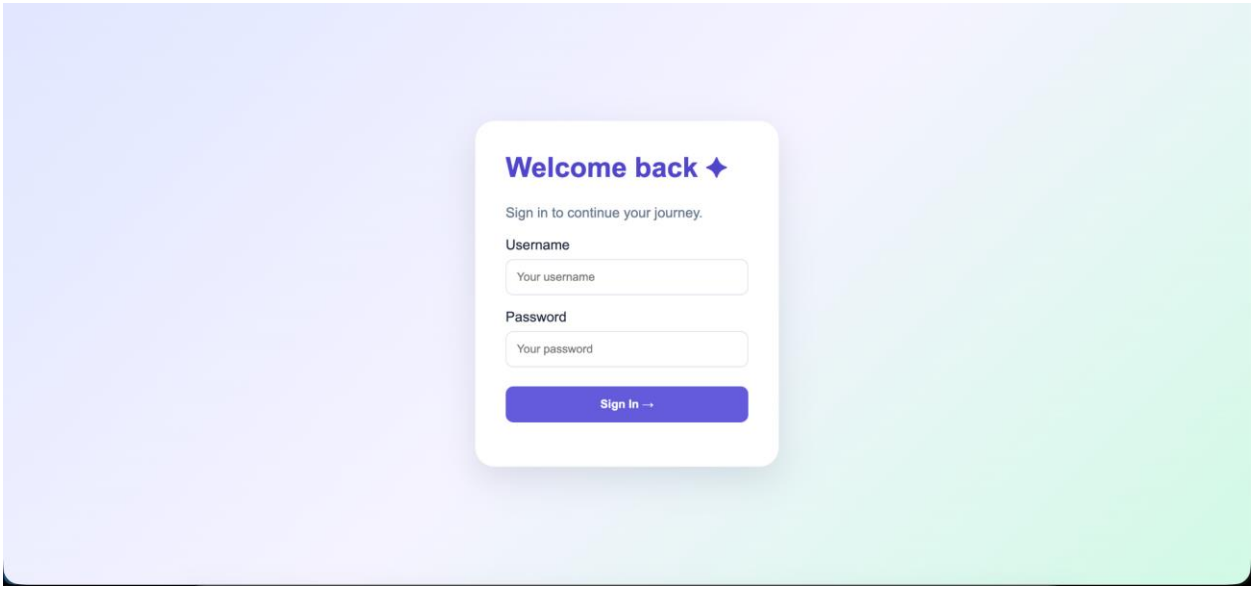


Figure 3. Question 3 login page.

The success screen is produced after submitting a username in the browser demo. A separate screenshot of that screen was not included among the three images in this final upload.

Conclusion

The three solutions demonstrate semantic HTML, corrected document structure, responsive design, client-side interaction and the HTTP concepts behind form submission. The HTML-only demonstrations do not persist data or send real server requests unless connected to a backend.

CODE OF CONDUCT

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Thorough, accurate understanding	Good understanding with minor gaps	Basic understanding	Major misconceptions
Explanation	25%	Detailed with examples	Clear explanation	Limited detail	Unclear explanation
Application	20%	Effective application	Minor errors	Limited application	Incorrect application
Organization	15%	Logical and structured	Mostly organized	Basic structure	Poor organization
Accuracy	10%	Accurate and complete	Mostly accurate	Partly correct	Major errors
Clarity	5%	Clear and professional	Understandable	Limited clarity	Difficult to follow